Current study

This study investigates the relationship between the aperiodic spectral exponent and local gamma phase-locking as measured by ITPC during 40-Hz ASSR in healthy participants. We analyzed MEG data from 15 participants during auditory stimulation.

Our analytical strategy utilizes Bayesian multilevel modeling to examine how spectral slope might be related to ITPC measured across brain regions while appropriately handling the multiple comparisons inherent in whole-brain analysis. This approach allows us to take into account region-specific effects while borrowing strength across regions through partial pooling, providing more robust and interpretable results than traditional mass-univariate approaches.

Understanding the relationship between these two putative biomarkers of local neural circuit function has important implications for their use in clinical and research contexts. If they reflect shared biological mechanisms, their combined examination may provide more sensitive and specific markers of local circuit integrity than either measure alone.

Definitive testing of biological relationship is not easy, hovever, we can partilly test methodological confound hypothesis with simulations of signals with verying paramenters. The slope-ITPC relationship may partially reflect methodological interdependencies in phase estimation, where steeper slopes (indicating stronger high-frequency attenuation) systematically bias ITPC measurements toward lower values by reducing the signal-to-noise ratio of oscillatory components relative to aperiodic background.

Participants and experimental design

This study analyzed a magnetoencephalography (MEG) dataset previously described by Pellegrino et al. (2019). The experimental procedures were approved by the Ethics Committee of the Province of Venice, and all participants provided written informed consent before participation.

Fifteen healthy, right-handed volunteers participated (12 females, 3 males; mean age = 28.8 $\pm$ 3 years, age range: 20-50 years). Participants were screened using the following exclusion criteria: (1) hearing impairment, (2) current use of medications affecting the central nervous system, (3) history of neuropsychiatric disorders, and (4) any other medical condition that might affect brain function.

The experiment used a within-subjects design to investigate cortical gamma-band synchronization in response to 40 Hz amplitude-modulated auditory stimuli. As shown in Figure 1, the protocol comprised two different sessions of bi-hemispheric tDCS stimulation (sham and real) conducted at least 20 hours apart indicated by condition (T): REAL or SHAM. The order of the tDCS stimulation (sham and real) was counterbalanced across subjects. Stimulation was applied to the left and right pericentral regions Figure 1 using standard protocols. The anode was positioned over the left hemisphere and the cathode over the right hemisphere, targeting auditory cortical regions. Real stimulation lasted 20 min with 20 s of fade-in and fade-out. Sham stimulation was performed with the same settings, but electrical current was delivered only for 20 s at the beginning and at the end of the tDCS session.

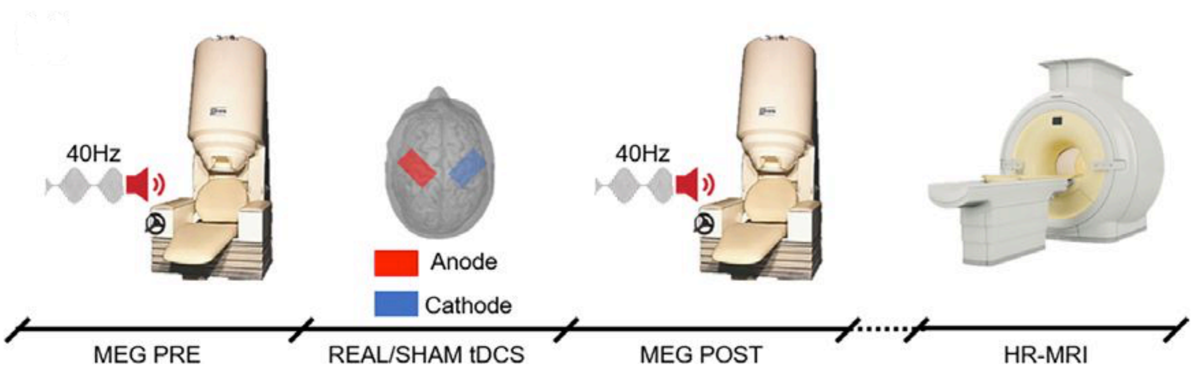
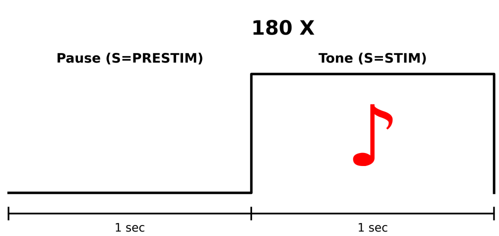
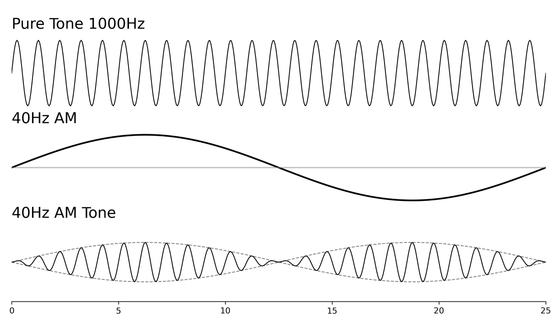


Figure 1: Design of the orginal experiment. Source: (Pellegrino et al. 2019)

Each session included two blocks corresponding to conditions related to the time (P) of applying tDCS stimulation: PRE (before tDCS) and POST (after tDCS). Figure 2a shows the structure of a paradigm used to induce 40 Hz synchrony during a single block. Each block consisted of 180 repetitions lasting 2 s each. Each trial included 1 s of pause (S=PRESTIM condition) followed by a 40 Hz amplitude-modulated (AM) tone (S=STIM condition), so that the stimulus onset asynchrony was 2 s. Each block lasted a total of 6 minutes (180 trials $\times$ 2 seconds). Figure 2b illustrates the auditory stimulus, an amplitude modulated tone with the following parameters: carrier frequency = 1,000 Hz and amplitude modulating frequency = 40 Hz.



(a) Structure of the trial.



(b) Carrier and amplitude modulation.

Figure 2: Structure of auditory stimulation. Adapted from Pellegrino et al. (2019).

MEG Data acquisition and preprocessing

MEG scans were performed at the MEGLab of the IRCCS San Camillo Hospital in Venice in a shielded room with a CTF-MEG system equipped with 275 gradiometers. As outlined in (Pellegrino et al. 2019) MEG data preprocessing included: third-order spatial gradient noise cancellation, downsampling to 600 Hz, signal space-separation. Heartbeat and eye-movement artefacts were cleaned with the SSP approach. MEG data was cut in 3 s epochs ranging from -1.5 s to $+1.5$ s around the 40 Hz sound onset. Epochs were visually inspected and those affected by artefacts were manually rejected. The available MRI scans were segmented and processed through Freesurfer software. The surface was divided into 15,000 vertices imported into Brainstorm toolbox and downsampled into 8.000 vertices. The coregistration between functional MEG data and anatomical MRI data was performed via the Brainstorm toolbox.

Data was bandpass filtered betweeen 2 and 80 Hz and then notch filtered for 50~Hz. The inverse problem for source estimation was solved with the whitened and depth-weighted linear L2-minimum norm estimate, with the dipole orientation constrained to be normal to the cortical surface. A common imaging kernel was computed and then applied to obtain single epoch cortical reconstructions. Data was projected into regions using Destrieux et al. (2010) atlas and exported from Brainstorm into MNE Python.

PSD and Aperiodic Parametrization

For every trial, the 1s second fragment before sound stimulation and 1s fragment during the sound stimulation was extracted. For those intervals power spectral density was computed using MNE Python using `welch` method and the median was taken across trials in a block (as is more roboust than just averaging). Then the slope and exponent of the spectrum were computed for each region of interest using `specparam` toolbox (Donoghue et al. 2020).

In the algorithm the power spectrum PSD is modeled as a combination of the aperiodic component AP and a sum of N oscillatory peaks modeled as a Gaussians:

$$P = AP + \sum_{n=0}^{N} G_n.$$

The component $AP(f)$ for frequency f can expressed by the formula:

$$AP(f) = b - \log(k + f^{\chi})$$

where b is the broadband offset, chi is the exponent and k is the knee parameter. The exponent χ is related to the negative slope of the power spectrum in log-log space, that is, the inverse relationship between power and frequency; The offset b describe the uniform shift in power across frequencies. The frequency range on which the algorithm was fitted span interval

[3, 45]. The parameters with which data were analysed are the following hyperparameters : `peak_width_limits` =[1, 8], `min_peak_height`=0.15, `peak_threshold`=2.0.

Estimation of synchrony via inter-trial phase coherence (ITPC)

Spectral modulations and phase consistency of ASSR in MEG and EEG data are highly reproducible, providing support for other research (Tan, Gross, and Uhlhaas 2015; Pellegrino et al. 2019). ASSR are usually investigated by looking at the inter-trial phase coherence (ITPC) . The inter-trial phase coherence (ITPC) is defined using the length of the mean phase angle,

$$\mathrm{ITPC} = \left| \frac{1}{N} \sum_{k=1}^{N} e^{i \cdot \phi_k} \right|,$$

where N represents the number of trials, $e^{i\phi_k}$ is complex polar representation the phase angle ϕ of the trial k. ITPC is an measure of phase consistency over trials, can take values between 0 and 1. The higher is ITPC, the higher is the 40 Hz gamma synchronization

The ITPC was computed for 1s second pre-stimulus period and 1s stimulation period via Time-Frequency Representation (TFR) using Morlet wavelets approach implemented in MNE Python (1.9) in the 39-41 Hz gamma band for each epoch.
Values of ITPC were extracted in the period of -500ms to - 200ms in the period before sound stimulation starts and 300 ms to 700 ms during the sound stimulation, counting on the time of brain reaching steady-state (Pellegrino et al. (2019)) and averaged in those periods.

Mapping of value distributions of variables of interest

To examine potential differences in inter-trial phase coherence (ITPC) across experimental conditions, we conducted a Bayesian multilevel analysis using the brms package (Bürkner, 2017) in R (R version `4.5.1` and brms version `2.22`).

First part we used data from the PRE condition

We looked into simple model where and how much ASSR change ITPC. My assumption is that the effect of sound stimulation will be dependent both from the Subject and roi (Barr et al. (2013) argues for maximal structure of model being conservative)

$$ITPC \sim 1 + S + (1 + S|ROI) + (1 + S|Subject)$$

Similarly we do use similar structure for parameters of the slope - exponent and offset

$$exponent \sim 1 + S + (1 + S|ROI) + (1 + S|Subject)$$

$$offset \sim 1 + S + (1 + S|ROI) + (1 + S|Subject)$$

We decided to fit sigma in the models as well to capture the uncertainty more precisely - Different subjects may not only differ in their mean responses but also in their consistency and ifferent ROIs might have different signal-to-noise ratios.

Main Analysis

A hierarchical Bayesian model was implemented using the brms package to assess the relationship between ITPC and spectral exponent. The model was specified as:

$$itpc \sim 1 + exponent_z + (1 + exponent_z|roi) + (1 + exponent_z|subject)$$

This formula represents ITPC as a function of the standardized spectral exponent (exponent_z), while accounting for random intercepts and slopes for both brain regions (roi) and subjects. We employed Student's t-distribution as the likelihood function to accommodate potential outliers in the data. The model was fitted using Markov Chain Monte Carlo sampling with 4 chains, 4000 iterations per chain, and a 1000-iteration warm-up period. To improve convergence, we set the adapt_delta parameter to 0.95.

To interpret region-specific effects, we extracted random effects for each ROI using the extract_region_effects function. This allowed us to calculate:

Base ITPC values (intercept) for each brain region The effect of spectral exponent on ITPC in each region Posterior probabilities (P+) indicating the likelihood of positive effects 95% credible intervals for parameter estimates

We calculated predicted ITPC values at different levels of the spectral exponent (±1 standard deviation from the mean) to quantify effect sizes across regions. Brain regions were then ranked based on both the magnitude of the spectral exponent effect and the posterior probability of the effect being positive or negative. Results were visualized using brain plots to illustrate the spatial distribution of effects.

ITPC Comparison Between Experimental Conditions including tDCS

We constructed a hierarchical Bayesian model with ITPC as the dependent variable. Fixed effects included the main effects of measurement time point (P: pre vs. post), stimulation type (T: tDCS vs. sham), and their interaction (P×T). The model incorporated subject-specific and ROI-specific random intercepts and slopes for all fixed effects, allowing for individual variation in both baseline ITPC values and responses to the experimental manipulations. The model specification was:

$$itpc \sim\ 1 + P * T + (1 + P * T|roi) + (1 + P * T|subject)$$

We used a Student's t-distribution as the likelihood function to accommodate potential outliers in the data. The model was fitted using four Markov Chain Monte Carlo (MCMC) chains with 4,000 iterations each (including 1,000 warmup iterations), resulting in 12,000 post-warmup samples for posterior inference. To improve convergence, we set the adapt_delta parameter to 0.95.

For hypothesis testing, we examined the posterior distributions of the regression coefficients, focusing on the main effects of time point and stimulation type, as well as their interaction. We calculated probability values (P+) for each ROI-specific parameter, representing the probability of a positive effect when the mean was positive, or a negative effect when the mean was negative. Effects were considered to show strong evidence when P+ exceeded 0.95 (strong evidence for a positive effect) or fell below 0.05 (strong evidence for a negative effect).

Simulations

Descriptive Statistics and Data Quality

Before examining the relationship between spectral exponent and ITPC, we first verified that our experimental manipulations (tDCS stimulation) did not systematically alter ITPC measurements, allowing us to pool data across conditions for our primary analysis

Results

ITPC

First we tested if the conditions PRE in real and sham conditions aren't different. No region has P+ values bigger than 0.90 or smaller than 0.10 so we concluded that these condition's aren' different.

Simple visualization of the results ITPC results are presented in the Figure 3.

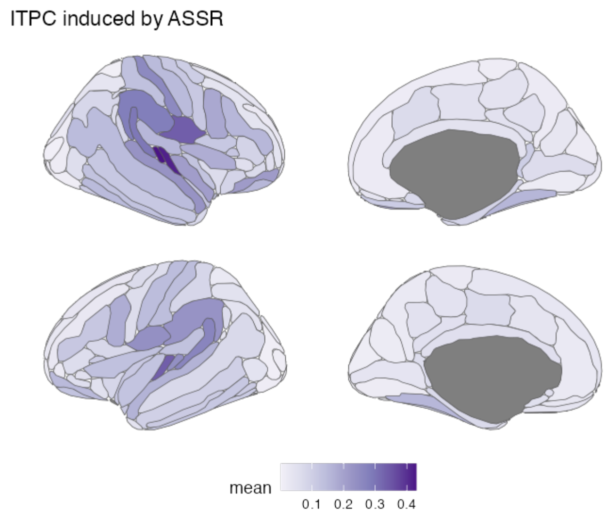


Figure 3: Z-normalized values of ITPC (for PRE Sham and PRE real conditions)

The region with the biggest response values were The P+ (Part of posterior distrubution that is bigger than 0) is presented in the table

We also computed the map o difference Pre sham/real tDCS effects and then the Real versus Sham stimulation were contrasted (computed the difference) using Bayesian Multilevel Modelling, but no region has P+ suggesting effect of TDCS.

Specparam parameters

We tested effects of stimulation on exponent and we didnt find any difference. By meanigful differnce we count those with P+ > 0.975 or 0.025. Below we present exponent values as mapped per region

Mean values of parameters describing the 1/f part od spectral power across different regions (right hemisphere).

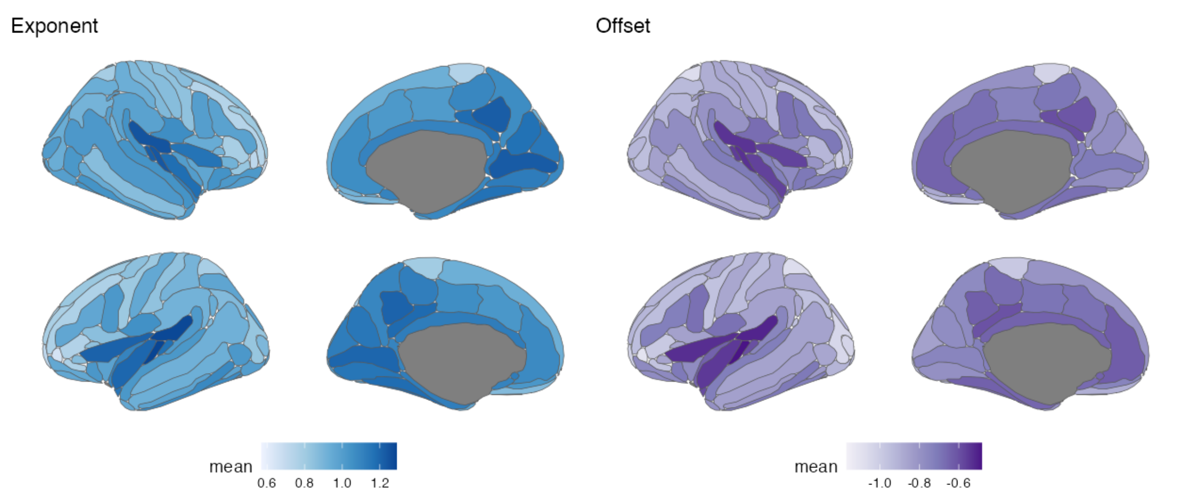


Figure 4: Mean values of exponent and offset averaged for the trials before tDCS stimulation and before sound stimulation

Evidence of Three Explanatory Models for the ITPC

- The exponent and the offset of the spectrum are different across the brain (Box 1); The parametrization of the slope of the aperiodic component (exponent) might be affected by neural entrainment and increased power in gamma frequency
- Strong effect of the stimulus (S) on the ITPC (contrast baseline-stimulus), in areas like the auditory regions, but also in temporal lobe, inferior part of primary sensory-motor regions and inferior frontal regions; negative values indicate higher ITPC during stimulation.
- Model with two explanatory variables

  $$ITP \sim Exponent + Stimulus$$

  has strong evidence in the same regions, both when considering exponent and stimulus as explanatory variables, with the effects having opposite signs, on the same regions.
- No particular evidence was found for the model #strong[Exponent ~ Stimulus];, namely the stimulation did not influence the exponent per se.
- No significant evidence was found for any of the models involving the Offset either as a predicted variable, and as a predictor, suggesting a marginal role for it in the auditory entrainment.

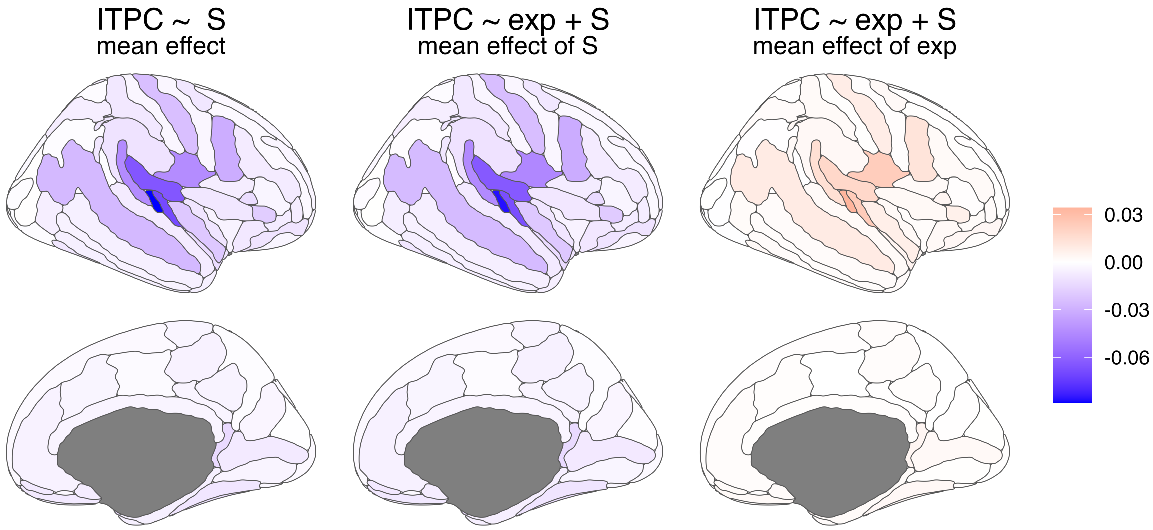


Figure 5: Positive values indicate stronger modulation at baseline, negative values indicate stronger modulation during stimulation.

ITPC including tDCS

Additionally, we extracted the ROI-specific coefficients to examine regional variation in the effects. This allowed us to identify potential regions showing differential responses to the stimulation conditions. The preliminary analysis revealed no significant differences in ITPC between pre- and post-stimulation timepoints, nor between tDCS and sham conditions. The interaction between timepoint and stimulation type also showed no significant effect. These findings suggest that ITPC measurements can be appropriately pooled across these conditions in subsequent analyses, as the experimental manipulations did not substantially alter the phase coherence measures.

References

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